LIMITED COORDINATION MILITARY SPECIFICATION

SAW SYSTEM, BONE, ELECTRIC

This specification is approved for use within the Department of the Defense and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers an electrical saw system suitable for sawing all small bones during surgical procedures.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications and standards. The following specifications and standards form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of these documents shall be those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement theretom cited in the solicitation.

SPECIFICATION

MILITARY

MIL-D-42048 - Deployable Medical Systems, General Requirements For DPSC - DEPMEDS - AT(DM).

STANDARDS

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Headquarters, Defense Personnel Support Center, ATTN: Directorate of Medical Materiel, DPSC-RST, 2800 S. 20th Street, Philadelphia, PA 19101, by using self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC 6515

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MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.

2.1.2 Drawing. The following drawing forms a part of this specification to the extent specified herein. Unless otherwise specified, the issue shall be those in effect on the date of solicitation.

DEFENSE PERSONNEL SUPPORT CENTER (DPSC) DRAWING

22403 - BLADES, RECIPROCATING, OSCILLATING, SAGITTAL

(Copies of specification, standards and drawings required by contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting activity).

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of the documents which are DoD adopted shall be those listed in the issue of the DoDISS specified in the solicitation. Unless otherwise specified, the issues of documents not listed in the DoDISS shall be the issue of the nongovernment documents which is current on the date of the solicitation.

UNDERWRITERS LABORATORIES (UL)

UL 544 - Standard for Safety, Medical and Dental Equipment.

(Application for copies should be addressed to Underwriters Laboratories, 333 Pfingsten Road, Northbrook, IL 60062).

UNITED STATES PHARMACOPEIAL CONVENTION, INC

Pharmacopeia of the United States (USP)

(Application for copies should be addressed to the Mack Publishing Company, Easton, PA 18042).

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

ASTM F899 - Stainless Steel Billet, Bar and Wire for Surgical Instruments.

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(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103).

(Nongovernment standards and other publications are normally available from the organizations which prepare or which distribute the documents. These documents also may be available in or through libraries or other informational services).

2.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence. Nothing in this specification, however, shall supersede applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 First article. When specified in the contract or purchase order, a sample shall be subjected to the first article inspection (see 4.1.2.1 and 6.3).

3.2 Materials.

3.2.1 Saw blades. Blades shall be fabricated from 420 stainless steel, in accordance with ASTM F899.

3.2.2 Electrical components. Electrical components shall be suitably insulated to prevent shock to the operator and to the patient.

3.2.3 Other components. Other components shall be made of materials suitable for repeated usage without deteriorating.

3.3 Components. The saw system shall consist of the following components and supplied in their respective quantities.

Component	Quantity
Power console	1
Footswitch	1
Motor	1
Reciprocating, saw	1
Blade, reciprocating, 6 mm x 15 mm	2
Blade, reciprocating, 6 mm x 22 mm	2
Blade, reciprocating, 6 mm x 28 mm	2

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Component	Quantity
Oscillating Saw	1
Blade, oscillating, 9.5 mm x 33 mm	10
Blade, oscillating, 9.5 mm x 18.5 mm	10
Blade, oscillating, 5.5 mm x 25.5 mm	10
Sagittal Saw	1
Blade, sagittal, 14 mm x 41 mm	10
Blade, sagittal, 6.5 mm x 25.5 mm	10
Blade, sagittal, 9.5 mm x 25.5 mm	10
Sterilization case	1
Allen wrench	1
Silicone lubricant	1

3.4 Style and design. The following style and design characteristics shall apply.

3.4.1 Power console. The console shall be a solid state, electric power supply for supplying variable direct current (DC) to the motor to operate reciprocating, oscillating and sagittal saws used for small bone surgery. The console shall weigh no more than 12 lbs., and be no more than 5 inches high by 11 inches wide by 13 inches deep. The console shall have manually operated rheostat for controlling the motor speed. The output shall be controlled by a variable speed footswitch. The console shall have a current overload protection feature to protect the motor windings switch and a female adapter for locking the motor power cable. The front of the console shall have a switch for changing speed control from the console to the footswitch. A power on/off indicator shall light red when activated. A footswitch indicator shall light green when the footswitch is activated. The console shall have an adapter to lock a variable speed footswitch cable. The console enclosure shall be aluminum.

3.4.2 Footswitch. The footswitch shall control motor speed. The footswitch shall have an insulated cable measuring 8 feet plus or minus 3 inches. An adapter on one end of the cable shall be designed to connect the cable to the console. The footswitch shall control the the speed of the motor by application and release of foot pressure. The maximum speed available by the footswitch shall be limited by the console speed setting. The footswitch shall weigh no more than 2 lbs.

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3.4.3 Motor. The motor shall power each electrical saw by means of an attached power cable from the console. The motor shall have a free running speed from 0 to 30,000 RPM. The motor shall operate on a variable voltage of 15 to 42 volts DC. The motor shall weigh no more than 7 ounces and measure 4-3/16 ± 3/16 inches long by 7/8 ± 1/8 inches diameter. The motor shall be permanently sealed within an aluminum housing and shall not require lubrication. The distal end of the motor shall have a rapid locking coupler for attaching a saw. The motor shall be able to be sterilized by 275 degree Fahrenheit (F) steam.

3.4.4 Reciprocation saw. The reciprocating saw shall be straight handled, pencil-like design compatible with the reciprocating saw blades included with the system. The saw shall provide 0 to 20,000 reciprocations per minute, a stroke length of 0.11 inches, and accept saw blades and rasps with 1/8 inch diameters. The saw shall be permanently sealed, measure not more than 6 inches long by 1 inch wide, and weigh not more than 7 ounces. The proximal end shall have integral metal projections for adapting securely into the coupler on the distal end of the motor. The distal end shall have a knurled spring lock mechanism allowing insertion and removal of blades and rasps without the aid of tools. The saw shall be able to be sterilized by 275 degree F steam. The metal portions of the saw shall be a suitable corrosion-resistant steel, properly hardened and passivated.

3.4.5 Reciprocating saw blades. The reciprocating saw blades shall securely lock into the reciprocating saw. The reciprocating blades shall be as shown in DPSC Drawing No. 22403.

3.4.6 Oscillating saw. The oscillating saw shall be of straight design and compatible with the oscillating saw blades included with the system. The saw shall provide 0 to 30,000 oscillations per minute and permit a 4 degree blade excursion. The saw shall be permanently sealed, measure not more than 3-1/4 inches long, and weigh not more than 5 ounces. The proximal end shall have integral metal projections for adapting securely into the coupler on the distal end of the motor. The distal end shall seat the oscillating blades perpendicular to the handle by means of an Allen wrench. The saw shall be able to be sterilized by 275 degree F steam. The metal portions of the saw shall be a suitable corrosion-resistant steel, properly hardened and passivated.

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3.4.7 Oscillating saw blades. The oscillating saw blades shall have an elongated U-notch cut-out of the blade for attachment to the distal end of the saw. The blades shall be sterile in accordance with the U.S.P.. The blades shall be individually wrapped in a peel-open envelope, 5 blades to a package. Each package of blades shall contain the legend (or similar) "Sterile unless opened or damaged". The oscillating blades shall be as shown in DPSC Drawing No. 22403.

3.4.8 Sagittal saw. The sagittal saw shall be of straight design and be compatible with the sagittal saw blades included with the system. The saw shall provide 0 to 30,000 oscillations per minute and permit a 3 degree blade excursion. The saw shall be permanently sealed, measure not more than 3-1/4 inches, and weight not more than 5 ounces. The proximal end shall have integral metal projections for adapting securely into the coupler on the distal end of the motor. The distal end shall seat the elongated U-notched of the sagittal saw blades by means of an Allen wrench. The sagittal saw blades shall permit blade rotation from 0 to 220 degrees. The saw shall be able to be sterilized by 275 degree F steam. The metal portions of the saw shall be a suitable corrosion-resistant steel, properly hardened and passivated.

3.4.9 Sagittal saw blades. The sagittal saw blades shall have an elongated U-notch cut-out of the blades for attachment to the distal end of the saw. The blades shall be sterile in accordance with the U.S.P.. The blades shall be individually wrapped in a peel-open envelope, 5 blades to a package. Each package of blades shall contain the legend (or similar) "Sterile unless opened or damaged". The sagittal blades shall be as shown in DPSC Drawing No. 22403.

3.4.10 Sterilization case. The sterilization case shall be fabricated of a suitable corrosion-resistant steel, properly hardened and passivated, with at least 252 holes for steam or ethylene oxide penetration. The holes shall measure 5 ± .05 millimeters in diameter. The case shall measure 10 ± 1 inches wide, 7 ± 1 inches deep, and 3-1/2 ± 1/2 inches high. The case shall have a hinged top and a flat closure mechanism for securing the lid. The exterior surfaces shall be flat, without a handle, facilitating stacking of cases.

3.4.11 Allen wrench. The Allen wrench shall be a 1/8 inch hex head Allen wrench with knurled handle suitable for locking and unlocking blades. The wrench shall measure 5 ± 1 inches in overall length with a handle length of 3 ± 1/2 inches. The wrench shall be fabricated of a suitable corrosion-resistant steel, properly hardened and passivated.

3.4.12 Silicone lubricant. The silicone lubricant shall be a nonflamable, aerosol spray for lubrication of the saws. The lubricant shall be nontoxic to human skin and mucosa, and not inhibit sterilization procedures of the saws.

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3.5 Finish. The components shall be fabricated from materials which are resistant to corrosion or which are treated to resist corrosion. Finish shall be free of gaps, scratches, chips, and other imperfections.

3.6 Electrical characteristics. The saw system shall be listed by Underwriters Laboratories or other acceptable testing agency. Electrical characteristics shall be consistent with the requirements of MIL-D-42048.

3.6.1 Dielectric strength. When tested as specified in 4.4.1, the saw system shall meet the requirements of U.L. 544 for dielectric strength.

3.6.2 Current leakage. When tested as specified in 4.4.2, the saw system shall not exceed the safe current leakage limit of U.L. 544.

3.7 Motor temperature rise. When tested as specified in 4.4.3, the temperature of the motor shall not rise more than 40 degrees F.

3.8 Autoclave capability. When tested as specified in 4.4.4, the motor, saws, blades, sterilization case, and Allen wrench shall show no evidence of corrosion or operational defects.

3.9 Motor torque. When tested as specified in 4.4.5, the motor torque shall not exceed 6 inch-ounces.

3.10 Reusable container. A reusable container shall be provided in accordance with the requirements of MIL-D-42048 (see paragraph 3.6).

3.11 Instructions. The saw system shall be supplied with two copies of instructions for operations and maintenance of the components.

3.12 Workmanship. The saw system shall be free from defects which detract from its appearance or impair its serviceability.

3.13 Units. Each (EA). One saw system, as specified, constitutes one unit.

3.14 Provisioning. Shall be provisioned in accordance with the appendix of specification DPSC-DEPMEDS-AT(DM).

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4. QUALITY ASSURANCE PROVISIONS.

4.1 Responsibility for inspection. Unless otherwise specified in the contract, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1 Responsibility of compliance. All items must meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of assuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling in quality conformance does not authorize submission of known defective material, either indicated or actual, or does it commit the Government to acceptance of defective material.

4.1.2 Inspection. Inspection, as used in this specification, is defined as both examination (such as visual and auditory investigation without the use of special laboratory appliances or procedures) and testing (determination by technical means of physical and chemical properties) of the item.

4.1.2.1 First article inspection. First article inspection shall be performed on the saw system when a first article sample is required. This inspection shall include examination and tests.

4.1.3 Records. Records of examinations and tests performed by or for the contractor shall be maintained by the contractor and made available to the Government upon the Government's request, at any time, or from time to time, during the performance of the contract and for a period of three years after delivery of the supplies to which such records relate.

4.2 Sampling.

4.2.1 For examination. Sampling for examination shall be conducted in accordance with MIL-STD-105 and table I. The unit of product for sampling purposes shall be one saw system.

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TABLE I. Sampling for examination.

Type of defect	Inspection level	AQL (percent defects)
Major defects	II	1.0
Minor	II	4.0

4.2.2 For tests. Sampling for tests shall be conducted in accordance with MIL-STD-105 and as indicated in Table II. The unit of product for test purposes shall be one saw system.

TABLE II. Sampling for tests.

Characteristic	Requirement	Test Procedure	Inspection Level	AQL(percent defective)
Dielectric strength	3.6.1	4.4.1	S-1	*
Current leakage	3.6.2	4.4.2	S-1	*
Motor temperature rise	3.7	4.4.3	S-1	*
Autoclave capability	3.8	4.4.4	S-1	*
Motor torque	3.9	4.4.5	S-1	*

* Acceptance number is zero.

4.3 Examination. The saw system shall be examined to determine compliance with the requirements of this specification in accordance with the classification of defects in Table II.

TABLE II. Categories and Defects.

Categories	Defects
Major	
101	Evidence of Underwriters' Laboratories (or other acceptable testing agency) listing not obtained.
102	Power console switches, lights, or rheostat inoperative.
103	Leakage of lubricant from hand motor.
104	Hand motor cable not mating properly to power console adapter.
105	Footswitch cable not mating properly to power console adapter.
106	Difficulty in attaching saws to hand motor.

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TABLE II. Categories and Defects. (con't)

Categories	Defects
Minor	
201	Hand motor and saws not free from burs, sharp edges, and cracks.
202	Electric power cord, footswitch cable, or hand motor cable length not as specified.
203	Improper identification, labels missing.

4.4 Test. Tests shall be conducted to determine compliance with this specification.

4.4.1 Dielectric strength. The saw system shall be tested for dielectric strength as specified in U.L. 544.

4.4.2 Current leakage. The saw system shall be tested for current leakage as specified in U.L. 544.

4.4.3 Motor temperature rise. The motor shall be operated for 10 minutes at full speed at an ambrant temperature of 72 ± 5 degrees F.

4.4.4 Autoclave capability. The motor, saws, blades, sterilization case, and Allen wrench shall be steam autoclaved at 270 ± 20 minutes.

4.4.5 Motor torque. The motor shall be checked for torque while running at 23,000 ± 500 RPM.

5. PACKAGING

5.1 General. The complete requirements for packaging shall consist of this specification and the latest issue of Military Specification MIL-D-42048 except where otherwise indicated.

6. NOTES

6.1 Intended use. The saw system shall be suitable for sawing all small bones during surgical procedures.

6.2 Ordering data.

6.2.1 Acquisition requirements. Acquisition documents should specify the following:

(a) Title, number, and date of specification.

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(b) Selection of applicable levels of packaging and packing (see 5.1 and 5.2).
(c) Applicable National StocK Number.

6.3 First article. When a first inspection is required, the item should be (a first article sample), (a first production item), (a sample selected from the first 5 production items) (or it may be a standard production item from the contractor's current inventory as specified in 4.1.2). The first article should consist of one unit. The contracting officer should include specific instructions in acquisition documents regarding arrangements for examinations, approval of first article test results and disposition of first article. Invitations for bids should provide that the Government reserves the right to waive the requirements for samples for first article inspection to those bidders offering a product which has been previously acquired or tested by the Government, and that bidders offering such products, who wish to rely on such production or test, must furnish evidence with the bid that prior Government approval is presently appropriate for the pending contract.

6.4 National Stock Number coverage. This specification covers the following National Stock Number:

National Stock Number	Item Identification
6515-01-206-5965	Saw System, Electric For Small Bones

6.5 Subject term (key word) listing.

Saw
Reciprocating
Oscillating
Sagittal
Dielectric strength
Current leakage

MILITARY INTERESTS:

Custodians:

Army - MD
Navy - MS
Air Force - 03

Preparing Activity:

DoD - MB

Agent:

DLA -DM

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CIVIL AGENCY COORDINATING ACTIVITIES:

VA - OSS
PHS
FDA - MPQAS

Project No. 6515-4305

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-S-42056 (DM)	SAW SYSTEM, BONE, ELECTRIC

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR ☐ USER ☐ MANUFACTURER ☐ OTHER *(Specify)*: _____
b. ADDRESS *(Street, City, State, ZIP Code)*	

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

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